



Extrapolating a sequence

1 Which of these are triangular numbers? Tick **3** correct answers

☐ 1☐ 8☐ 12☐ 15☐ 21

2 Which of these are prime numbers? Tick **3** correct answers

☐ 1☐ 2☐ 11☐ 13☐ 15

3 7, 8, 15, 23, 38 ... are the first five terms of a sequence. Which of these could be a rule for the sequence? Tick **1** correct answer

☐ Add 1, add 3, add 5 and so on, differences increasing by 2 each time.☐ Add 1, add 7, add 13 and so on, differences increasing by 6 each time.☐ Add the two previous terms to get the next term.☐ Multiply the two previous terms to get the next term.☐ Multiply the previous term by 2 and subtract 1 to get the next term.

4 A linear sequence starts -4, 6, The next term is _____. Fill in the blank



5 A geometric sequence starts -4, 6, What is the next term? Tick **1** correct answer

☐ - 12

☐ - 9

☐ 4

☐ 9

☐ 12

6 Which of these could be the first 5 terms in a sequence with a common second difference? Tick **2** correct answers

☐ 27, 64, 125, 216, 343, ...

☐ 28, 36, 45, 55, 66, ...

☐ 30, 52, 76, 98, 122, ...

☐ 32, 48, 72, 108, 162, ...

☐ 36, 49, 64, 81, 100, ...